THRESHOLD DISCREPANCY BARRIERS FOR DEPTH-THREE SCALAR PLANTED-CLIQUE AMP SCHEDULES

ARGUS OPEN RESEARCH PROJECT

Abstract. For the planted-clique recovery problem at clique size $k = \lceil \sqrt{n}/\log n \rceil$, we prove two exact limitations on frozen scalar approximate message passing (AMP) / cavity transfer schedules: (1) uniform positive-semidefinite domination via the collision Gram matrix alone fails at reachable parity states; and (2) every admissible rooted-C_4, depth-three scalar score has root-in/root-out threshold discrepancy $O(1/\log n + n^{-1/2})$, preventing sub-$\sqrt{n}$ candidate isolation.

1. Main Barriers

Theorem 1.1. *At $k = \lceil \sqrt{n}/\log n \rceil$ in $G(n, 1/2, k)$:*

(1) *Under admissible Lipschitz scalar denoisers, a reachable parity configuration sends a zero root-out shifted-Gram block to a nonzero rank-one all-ones block, ruling out uniform PSD pullback domination by the collision Gram matrix.*

(2) *For the rooted-C_4 depth-three scalar message schedule, signed diagonal cancellation incurs a loss of $k - 1$ and bounds the root-in versus root-out score discrepancy by $O(1/\log n + n^{-1/2}) = o(1)$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 2800203 in the domain of high-dimensional probability and statistical inference follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped impossibility theorem* deciding the scope: collision-Gram PSD pullback barrier and O(1/log n + n^{}{-1/2}) threshold discrepancy for depth-3 rooted-C4 scalar schedules.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- higher-depth denoisers and general sub-sqrt(n) planted-clique algorithms.

References

[1] Primary problem list and source literature for Result 2800203 (high-dimensional probability and statistical inference), archived with exact theorem locators and verification certificates in the accompanying repository package.

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 60B20; Secondary 68Q87, 62H30.